

05-SEP-2023

SUSPECTED UNEXPECTED SERIOUS ADVERSE REACTION (SUSAR) REPORT

SUSAR Report for Nivolumab (BMS-936558), # 2023-075847

Reference Document: Investigator brochure, Version 21 (01-Jul-2022)

Protocol: CA224-073

Dear Investigator:

As the sponsor of clinical studies with the investigational medicinal product (IMP) specified above, Bristol-Myers Squibb Research & Development is issuing this safety report. The details of this adverse event/safety issue are being reported to you in compliance with regulations.

For Suspected Unexpected Serious Adverse Reactions (SUSARs) occurring in blinded clinical trials, study treatments will remain blinded in reports sent to investigators.

The attached report does not reflect a conclusion or constitute an admission by the sponsor that the IMP caused or contributed to this adverse event/safety issue.

Your Institutional Review Board (IRB)/Ethics Committee (EC) must receive a copy of this SUSAR as per applicable regulation and local IRB/EC policy. Please consult the clinical trial personnel or documents/manuals regarding your responsibility for submitting SUSARs to your IRB/EC. Please also review your IRB/EC procedures relating to adverse events/safety issues that may necessitate changes to the Informed Consent Form.

Please include this SUSAR with all copies of the Reference Document.

We thank you in advance for your prompt attention to this information.

Krisztina Debreczeni, MD
Head, Medical Safety Review,
Worldwide Patient Safety

Attachment: CIOMS Report

SUSPECT ADVERSE REACTION REPORT													
	2023-075847(2)												

I. REACTION INFORMATION

1. PATIENT INITIALS (first, last)	1a. COUNTRY	2. DATE OF BIRTH			2a. AGE Years	3. SEX	4-6 REACTION ONSET			8-12 CHECK ALL APPROPRIATE TO ADVERSE REACTION
	MEXICO	Day	Month	Year	77	Female	Day	Month	Year	
							25	May	2023	
7+13 DESCRIBE REACTION(S) (including relevant tests/lab data) 1) Septic shock (English) MedDRA Version : v.26.0 SEPTIC SHOCK (Septic shock (10040070), Septic shock (10040070)) (27/May/2023 -) - Fatal 2) UROSEPSIS (English) UROSEPSIS (Urosepsis (10048709), Urosepsis (10048709)) (25/May/2023 -) - Recovering/Resolving										☒ PATIENT DIED ☒ LIFE THREATENING ☒ INVOLVED OR PROLONGED INPATIENT HOSPITALIZATION ☐ RESULTS IN PERSISTENCE OR SIGNIFICANT DISABILITY/INCAPACITY ☐ CONGENITAL ANOMALY ☒ OTHER MEDICALLY IMPORTANT CONDITION
Cont..										

II. SUSPECT DRUG(S) INFORMATION

14. SUSPECT DRUG(S) (include generic name)		20. DID EVENT ABATE AFTER STOPPING DRUG?
1) NIVOLUMAB (NIVOLUMAB) (NIVOLUMAB) (Suspect) (Verum) (Solution for injection) (Unknown)		
15. DAILY DOSE(S) (480 milligram(s))		☐ YES ☐ NO ☒ NA
16. ROUTE(S) OF ADMINISTRATION 1) Intravenous (not otherwise specified)		21. DID EVENT REAPPEAR AFTER REINTRODUCTION ☐ YES ☐ NO ☒ NA (NA : Not Applicable)
17. INDICATION(S) FOR USE 1) Hepatocellular carcinoma [10073071 - Hepatocellular carcinoma]		
18. THERAPY DATE(S) (from/to)	19. THERAPY DURATION	
1) (12/Jul/2022 - 09/May/2023)	1) 302 Days	

III. CONCOMITANT DRUG(S) AND HISTORY

22. CONCOMITANT DRUG(S) AND DATES OF ADMINISTRATION (exclude those used to treat reaction)	Cont..
1) CALCIUM CASEINATE (CALCIUM CASEINATE) (12/Sep/2022 - Unknown)	
23. OTHER RELEVANT HISTORY (e.g. diagnostics, allergies, pregnancy with last month of period, etc.)	Cont..
MedDRA Version : v.26.0 1) PYELONEPHRITIS (10037596, Pyelonephritis) (22/Feb/2023 - 13/Mar/2023) (Continuing: No)	

IV. MANUFACTURER INFORMATION

24a. NAME AND ADDRESS OF MANUFACTURER Name : BMS UNITED STATES OF AMERICA		Study Information Study Name: A Phase 2, Randomized, Open-label (Cont.) EudraCT Number: 2018-003151-38 Protocol No.: CA224-073 Center No.: 0106 Subject Id : 00312
24. REPORT NULLIFIED ☐ YES ☐ NO	24b. MFR CONTROL NO. 2023-075847(2)	
24c. DATE RECEIVED BY MANUFACTURER 01/Sep/2023	24d. REPORT SOURCE ☒ STUDY ☐ LITERATURE ☒ HEALTH PROFESSIONAL	
DATE OF THIS REPORT 05/Sep/2023	25a. REPORT TYPE ☐ INITIAL ☒ FOLLOWUP	

= Continuation attached sheet(s)..

Continuation Sheet for CIOMS report

7+13 DESCRIBE REACTION(S) (including relevant tests/lab data) (Continuation...)

Event Description :

Protocol title (CA224-073): A Phase 2, Randomized, Open-label Study of Relatlimab in Combination with Nivolumab in Participants with Advanced Hepatocellular Carcinoma who are Naive to IO Therapy but Progressed on Tyrosine Kinase Inhibitors (RELATIVITY-073)

PID: CA224-073-0106-00312 / 2023-075847 / Mexico

Reportable SAEs: Fatal septic shock and life-threatening (LT) urosepsis- SUSAR for nivolumab

A clinical investigator reported that a 77-year-old American Indian female patient was hospitalized due to CTC grade 4 LT urosepsis, approximately eleven months after the initiation of nivolumab study therapy for hepatocellular carcinoma. The patient received treatment for the event urosepsis. Two days later, the patient died due to septic shock. At the time of the patient's death, urosepsis was resolving.

The patient's medical history included hysterectomy, anemia and hernioplasty. She was non-alcohol user.

The patient's concomitant medications included calcium caseinate, nutrients NOS, ferrous sulfate, and prednisone (started 29-Dec-2022; ongoing at the time of urosepsis).

On 12-Jul-2022, open label study therapy was initiated with intravenous (IV) nivolumab (480 mg).

From 22-Feb-2023 to 13-Mar-2023, the patient had grade 2 non-serious event of pyelonephritis which was considered not related to the study therapy.

From 23-Mar-2023 to 12-Apr-2023, she had another episode of grade 2 non-serious event of pyelonephritis, which was considered not related to the study therapy.

On 06-May-2023, a computed tomography (CT) scan showed stable disease with no progression or new metastasis sites. Clinically, there was no evidence of disease progression.

On 09-May-2023, alpha-fetoprotein test showed minimal elevation compared to the baseline (results unspecified). On the same day, the patient received the last dose of study therapy prior to the onset of the events.

On 25-May-2023, the patient had generalized edema, and fever on two occasions (body temperature not reported). That same day, she had urosepsis.

On 26-May-2023, she was hospitalized due to severe dyspnea and drowsiness. Upon admission, she was noted to have polypnea, dehydration, hypoperfusion of the extremities, and grade 4 hypotension. Treatment was initiated with IV norepinephrine (15 mg), IV dobutamine (375 mg), IV hydrocortisone (200 mg) and vasopressin, as well as supplemental oxygen at a rate of 8 liters per minute.

On 27-May-2023, urinalysis showed a pH level of 6, leukocyte esterase was found to be 125 LEU/UL, proteins, glucose and erythrocytes were negative, nitrites were positive, and there were 20-30 leukocytes observed per field. Additionally, abundant bacteria were detected, which indicated a generalized urinary tract infection, and the patient was diagnosed with urosepsis. The laboratory test results showed hemoglobin 13.8 g/dL, platelet count 122,000/uL, leukocyte count 16,660/uL, neutrophil count 15,660/uL, urea 185 mg/dL, blood urea nitrogen (BUN) 86.4 mg/dL, creatinine 3.59 mg /dL, uric acid 16.6 mg/dL, glomerular filtration rate 11.62 ml/min/ 1.73 m², sodium 144.4 mEq/L, potassium 4.33 mEq/L, chloride 109.5 mEq/L, total bilirubin 5.8 mg/dL, direct bilirubin 4.4 mg/dL, indirect bilirubin 1.4 mg/dL, aspartate aminotransferase (AST) 535 U/L, alanine aminotransferase (ALT) 218 U/L, alkaline phosphatase (ALP) 593 U/L, gamma-glutamyl transferase (GGT) 517 U/L, lactate dehydrogenase ("DHL") 936 U/L, albumin 2 g/dl, and brain natriuretic peptide (BNP) 330.77pg/mL (reference ranges not provided). Antibiotic therapy was started with IV meropenem (1000 mg); however, the patient's condition deteriorated, and she died due to septic shock. It was unknown if an autopsy was performed.

At the time of death, treatment with norepinephrine, dobutamine, meropenem and hydrocortisone was ongoing, and urosepsis was resolving.

According to the investigator, fatal septic shock and LT urosepsis were non-immune mediated. Progressive disease was ruled out as the primary cause for urosepsis. There was no evidence of immune-related nephritis, along with no anatomic or mechanical etiologies that could explain the risk for fast evolution of urosepsis. The patient did not have any other medical condition that could affect the immune response, and despite the white blood cell response was apparently normal to infection, considering the rapid development of urinary sepsis, it appeared that the patient had immunosuppression. The patient was exposed to the checkpoint inhibitor study therapy for 11 cycles. She was not under treatment with any immunosuppressive or myelosuppressive agents in the past or prior to the onset of the events. No further assessments were performed as the patient requested voluntary discharge from the hospital. Based on all this and as Investigator's Brochure 21, Addendum 01 describes one case of urosepsis; urosepsis was considered related to the study therapy.

Supplemental information received on 01-Sep-2023 from the clinical investigator updated the dose of treatment medication hydrocortisone as 200 mg (previously reported as 400 ug), which was incorporated into the above narrative.

Investigator causality assessment: Fatal septic shock and LT urosepsis were related to nivolumab study therapy.

BMS causality assessment: Fatal septic shock and LT urosepsis were not related to nivolumab study therapy.

Continuation Sheet for CIOMS report

Expectedness Assessment:

Nivolumab (IB): Fatal septic shock and LT urosepsis were considered unlisted.

Company Remarks (Sender's Comments) :

This elderly patient was hospitalized due to urosepsis and subsequently died due to septic shock approximately nine months after the initiation and two and a half weeks after the last dose of nivolumab study therapy. The patient had two episodes of pyelonephritis prior to this hospitalization, that was considered not related to study therapy. Based on the available information, rapid evolution of urosepsis was the most probable cause for septic shock. Considering the infectious nature of the events (non-immune mediated), the reported events are considered not related to the study therapy. Concurrent long-term steroid treatment in this female patient with prior surgical history (hysterectomy/hernia repair) and pyelonephritis were significant risk factors for urosepsis.

Additional Information (Continuation...)

Laboratory Data :

See Event Description for additional lab results

Lab Result :

Test Name	Test Date	Test Result	Normal Value
ALBUMIN	27/May/2023	2 gram per decilitre	
ALKALINE PHOSPHATASE	27/May/2023	593 U/L	
ALPHA-FETOPROTEIN	09/May/2023		
ALT	27/May/2023	218 U/L	
AST	27/May/2023	535 U/L	
BNP	27/May/2023	330.77 picogram per millilitre	
BUN	27/May/2023	86.4 milligram per decilitre	
CHLORIDE	27/May/2023	109.5 milliequivalent per litre	
CREATININE	27/May/2023	3.59 milligram per decilitre	
CT SCAN	06/May/2023		
DHL	27/May/2023	936 U/L	
DIRECT BILIRUBIN	27/May/2023	4.4 milligram per decilitre	
GGT	27/May/2023	517 U/L	
GLOMERULAR FILTRATION RATE	27/May/2023	11.62 millilitre per minute per 1.73 square metre	
HB	27/May/2023	13.8 gram per decilitre	
HTO	27/May/2023	40.8 % percent	
INDIRECT BILIRUBIN	27/May/2023	1.4 milligram per decilitre	
LEUKOCYTES	27/May/2023	16660 /UL	

Continuation Sheet for CIOMS report

NEUTROPHILS	27/May/2023	15660 /UL	
PLATELET	27/May/2023	122000 /UL	
POTASSIUM	27/May/2023	4.33 milliequivalent per litre	
SODIUM	27/May/2023	144.4 milliequivalent per litre	
TOTAL BILIRUBIN	27/May/2023	5.8 milligram per decilitre	
UREA	27/May/2023	185 milligram per decilitre	
URIC ACID	27/May/2023	16.6 milligram per decilitre	
URINALYSIS	27/May/2023		

Test Result (Code) / Result Unstructured Data (free text) :

3) Test Name: ALPHA-FETOPROTEIN

Result Unstructured Data (free text) : MINIMAL ELEVATION COMPARED WITH BASELINE

Test Date: 09/May/2023

10) Test Name: CT SCAN

Result Unstructured Data (free text) : STABLE DISEASE WITH NO DATA ON PROGRESSION OR NEW METASTASIS SITES WERE REPORTED

Test Date: 06/May/2023

26) Test Name: URINALYSIS

Result Unstructured Data (free text) : PH 6, LEUKOCYTE ESTERASE 125 LEU/UL, PROTEINS NEGATIVE, GLUCOSE NEGATIVE, NITRITES POSITIVE, ERYTHROCYTES NEGATIVE, LEUKOCYTES 30-20 PER FIELD, ABUNDANT BACTERIA. SEPSIS OF URINARY ORIGIN WAS CONCLUDED

Test Date: 27/May/2023

14.SUSPECT DRUG(S) (Continuation...)

Product-Reaction Level

1) Drug : NIVOLUMAB (NIVOLUMAB)
 Active Substance : 1) NIVOLUMAB
 Coding Class : Verum
 Drug Characterization : Suspect
 Form of Admin : 1) Solution for injection
 Lot Number : 1) Unknown
 Daily Dose : (480 milligram(s))
 Route of Admin : 1) Intravenous (not otherwise specified)
 Indications : 1) Hepatocellular carcinoma [10073071 - Hepatocellular carcinoma]
 Therapy Dates : 1) From : 12/Jul/2022 To :09/May/2023
 Therapy Duration : 1) 302 Day
 Action(s) Taken With Drug : Not applicable

Causality

1) SEPTIC SHOCK (Septic shock - 10040070, Septic shock - 10040070)

Causality as per reporter : Related
 Causality as per Mfr : Not Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable

2) UROSEPSIS (Urosepsis - 10048709, Urosepsis - 10048709)

Causality as per reporter : Related
 Causality as per Mfr : Not Related
 DeChallenge : Not applicable
 ReChallenge : Not Applicable

Labeling :

1) SEPTIC SHOCK

Continuation Sheet for CIOMS report

CORE	UnLabeled
IB	UnLabeled
UNITED STATES OF AMERICA	UnLabeled

2) UROSEPSIS

CORE	UnLabeled
IB	UnLabeled
UNITED STATES OF AMERICA	UnLabeled

2) Drug : RELATLIMAB,NIVOLUMAB_JPN (RELATLIMAB,NIVOLUMAB_JPN)

Active Substance : 1) RELATLIMAB
2) NIVOLUMAB

Coding Class : Open-Label

Drug Characterization : Drug Not Administered

Form of Admin : 1) Injection

Lot Number : 1) Unknown

Daily Dose : (480 milligram(s))

Route of Admin : 1) Intravenous (not otherwise specified)

Indications : 1) Hepatocellular carcinoma [10073071 - Hepatocellular carcinoma]

Therapy Dates :

Therapy Duration : 1) 302 Day

Action(s) Taken With Drug : Not applicable

Causality

1) SEPTIC SHOCK (Septic shock - 10040070, Septic shock - 10040070)

Causality as per reporter : Related

Causality as per Mfr : Not Related

DeChallenge : Not applicable

ReChallenge : Not Applicable

2) UROSEPSIS (Urosepsis - 10048709, Urosepsis - 10048709)

Causality as per reporter : Related

Causality as per Mfr : Not Related

DeChallenge : Not applicable

ReChallenge : Not Applicable

Labeling :

1) SEPTIC SHOCK

IB	UnLabeled
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2) UROSEPSIS

IB	UnLabeled
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15. DAILY DOSE(S) (Continuation...)

Dosage Text :

Drug 1 :NIVOLUMAB

1) Recent dose on 09-May-2023 Not applicable to Urosepsis

Drug 2 :RELATLIMAB,NIVOLUMAB_JPN

1) Recent dose on 09-May-2023 Not applicable to Urosepsis

22.CONCOMITANT DRUG(S) (Continuation...)

1). Drug : CALCIUM CASEINATE

Active Substance : 1) CALCIUM CASEINATE

Form Strength :

Form of Admin : 1) Unknown

Route of Admin : 1) Oral

Indications : 1) worsening anemia [10054309 - Anemia aggravated]

Therapy Dates : 1) From : 12/Sep/2022 To : to Unknown

Dosage Text : 1) dose:2 unit nos

2). Drug : ENSURE

Active Substance : 1) NUTRIENTS NOS

Form Strength :

Form of Admin : 1) Unknown

Route of Admin : 1) Oral

Continuation Sheet for CIOMS report

Indications : 1) product used for unknown indication [10070592 - Product used for unknown indication]
 Therapy Dates : 1) From : 12/Sep/2022 To : to Unknown
 Dosage Text : 1) dose: 1 unit nos

3). Drug : FERROUS SULFATE
 Active Substance : 1) FERROUS SULFATE
 Form Strength :
 Form of Admin : 1) Unknown
 Daily Dose : 1) (200 milligram(s))
 Route of Admin : 1) Oral
 Indications : 1) worsening anemia [10054309 - Anemia aggravated]
 Therapy Dates : 1) From : 12/Sep/2022 To : to Unknown

4). Drug : PREDNISONE
 Active Substance : 1) PREDNISONE
 Form Strength :
 Form of Admin : 1) Unknown
 Daily Dose : 1) (5 milligram(s))
 Route of Admin : 1) Oral
 Indications : 1) Product used for unknown indication [10070592 - Product used for unknown indication]
 Therapy Dates : 1) From : 29/Dec/2022 To : to Unknown

23. OTHER RELEVANT HISTORY (Continuation...)

- 2) SUPPLEMENTAL OXYGEN (10050322 , Oxygen supplementation) (26/May/2023 - Unknown) (Continuing : NO)
 3) HYSTERECTOMY (10021151 , Hysterectomy) (Continuing : Unknown)
 4) HERNIOPLASTY (10088044 , Hernioplasty) (Continuing : Unknown)
 5) ANEMIA/WORSENING ANEMIA (10054309 , Anemia aggravated) (Continuing : Unknown)
 6) HEPATOCELLULAR CARCINOMA (10073071 , Hepatocellular carcinoma) (Continuing : YES)
 7) NON-ALCOHOLIC (10000340 , Abstains from alcohol) (Continuing : Unknown)

Study Information (Continuation...)

Study Name :

A Phase 2, Randomized, Open-label Study of Relatlimab in Combination with Nivolumab in Participants with Advanced Hepatocellular Carcinoma who are Naive to IO Therapy but Progressed on Tyrosine Kinase Inhibitors (RELATIVITY-073)

24. Other Study Registration Numbers

Registration Number

CA224-073

33478

2018-003151-38

Registration Country

TAIWAN, PROVINCE OF CHINA

KOREA, REPUBLIC OF

European Union

Primary Reporter:

